WHITE PAPER · HWACHEON MEGA 95 × MILITARY & DEFENSE

Hwacheon Mega 95 for Military & Defense

Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

Rattle & Hum · 16" OD × 120" L max turning · long-bed heavy-duty CNC lathe

Executive summary

Machine: Hwacheon Mega 95 (shop nickname: *Rattle & Hum*). long-bed heavy-duty CNC lathe at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 16" OD × 120" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Ten feet of turning length is a specialty capability. Long drive shafts, extended mandrels, rig-floor rollers, oil-and-gas structural rods — the parts that don't fit on a 4- or 6-foot lathe. Rattle & Hum lets us keep this class of work in-house instead of outsourcing the finish turning after roughing on a larger machine somewhere else.

Full spec table

Model	Hwacheon Mega 95 ("Rattle & Hum")
Type	Long-bed heavy-duty CNC turning center
Max turning capacity	16" outer diameter × 120" length
Bed length	10-foot working length for extended parts
Rigidity	Cast-iron construction; steady-rest support for long slender work
Tolerance capability	±0.0005" routine on critical features; straightness held over full length
Materials	Full oilfield alloy range — Inconel, Duplex, 17-4 PH, 4140/4340, Monel

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: ITAR compliance, security-sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design.

Typical Military & Defense parts we produce on Rattle & Hum

- Extended structural rods and shafts
- Long weapon-system components
- Extended defense-platform structural parts
- Long-form actuator components
- Sustainment parts requiring 8+ foot turning length

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

ITAR-controlled work under NDA with customer-specific document control — prints stay in-shop. Customer-specific quality plans, PPAP, and first-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we handle ITAR, we document properly, and we understand that a program-office audit is a real thing.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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